SCI225 · PATHOPHYSIOLOGY · WEEK 11 · NIGHTINGALE COLLEGE

The Renal & Urinary System

How the kidney filters, concentrates, and regulates — and what goes wrong: obstruction and stones, infection, acute kidney injury, chronic kidney disease, and the pediatric renal disorders.

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Module 11

Function · Alterations · Pediatric

9 topics + cheat sheet

SECTION 1

Structure & Function

How the nephron filters and concentrates, and how the kidney defends its filtration rate and controls blood pressure.

1

The Nephron, Filtration & GFR

▼

The **nephron** is the kidney's functional unit. Filter at the **glomerulus**, then fine-tune along the tubule.

Structure	Job
Glomerulus	**Filters** blood; the driving force is **capillary hydrostatic (blood) pressure** pushing fluid into the filtrate
Proximal tubule	**Reabsorbs most** of the filtered water, sodium, glucose, and nutrients (the bulk-reabsorption workhorse)
Loop of Henle	The **countercurrent system** builds the medullary concentration gradient → lets the kidney **concentrate urine**
Distal tubule / collecting duct	Final adjustments under **aldosterone** (Na/K) and **ADH** (water)

GFR and creatinine

Glomerular filtration rate (GFR) is how much blood the kidneys filter per minute — the key measure of kidney function. **Creatinine** is a waste product cleared by filtration, so when **GFR falls, creatinine rises** (it accumulates). A high plasma creatinine therefore means **reduced kidney filtration**.

> **Key idea:** filtration is driven by **blood (hydrostatic) pressure** in the glomerulus. The **proximal tubule reabsorbs the most**, and the **loop of Henle** (countercurrent) is what lets you make **concentrated** urine. Watch **creatinine** as the mirror of GFR — it goes **up** when filtration goes **down**.

PRACTICE

Q1. Which factor drives glomerular filtration?

A. Capillary hydrostatic (blood) pressure pushing fluid into the filtrate

B. Plasma oncotic pressure pulling fluid in

C. Tubular reabsorption

D. ADH secretion

A is correct. Glomerular capillary hydrostatic pressure is the main force pushing fluid across the filtration barrier into the tubule.

PRACTICE

Q2. A patient has an elevated plasma creatinine. What does this indicate?

A. GFR has declined, so creatinine accumulates

B. GFR has increased

C. Increased muscle hydration

D. Excess tubular secretion of creatinine

A is correct. Creatinine is cleared by filtration; when GFR falls it builds up in the blood — a marker of reduced kidney function.

PRACTICE

Q3. Which nephron structure contributes most to urine concentration?

A. The loop of Henle (countercurrent multiplier)

B. The glomerulus

C. The proximal tubule

D. The ureter

A is correct. The loop of Henle's countercurrent system builds the medullary osmotic gradient that allows water reabsorption and concentrated urine.

2 Hormonal Control & Autoregulation

▼

The kidney both **protects its own GFR** and **controls blood pressure/fluid balance**.

Autoregulation — keeping GFR steady

- The kidney adjusts the **afferent (incoming) and efferent (outgoing) arterioles** to hold GFR roughly constant despite changing blood pressure.
- **Sympathetic activation** constricts the **afferent arteriole** → less blood in → ↓ **GFR** (e.g., in shock, blood is shunted away from the kidney).

RAAS — the blood-pressure axis

- **Low blood pressure/flow** → the kidney releases **renin**.
- Renin → angiotensin II (**vasoconstriction**) + **aldosterone** (Na & water retention) → **raises blood volume and pressure**.

ADH — water control

ADH makes the collecting duct reabsorb **water** → concentrated urine, higher blood volume (links to Week 8's DI/SIADH).

> **Key idea: low BP → renin → RAAS → retain salt & water → BP up.** Autoregulation tweaks the two arterioles to stabilize GFR; strong **sympathetic drive constricts the afferent arteriole and drops GFR**. ADH adds water back when the body needs volume.

PRACTICE

Q1. A patient's blood pressure drops. Which mechanism explains increased renin release?

A. Reduced renal perfusion triggers renin to activate RAAS

B. High blood pressure stimulates renin

C. Excess sodium triggers renin

D. ADH suppresses renin

A is correct. Low pressure/flow to the kidney triggers renin release, launching RAAS to raise volume and pressure.

PRACTICE

Q2. Which physiologic response occurs with RAAS activation?

A. Increased sodium and water reabsorption, raising blood volume/pressure

B. Sodium loss and diuresis

C. Vasodilation lowering blood pressure

D. Decreased aldosterone

A is correct. Angiotensin II constricts vessels and aldosterone retains sodium and water — together raising blood volume and pressure.

PRACTICE

Q3. Sympathetic activation decreases afferent arteriole diameter. What results?

A. Reduced blood flow into the glomerulus and a lower GFR

B. Increased GFR

C. Increased renal blood flow

D. No change in filtration

A is correct. Constricting the afferent (incoming) arteriole reduces blood entering the glomerulus, lowering GFR — useful for redirecting blood during stress/shock.

SECTION 2

Alterations of Renal & Urinary Function

Obstruction and stones, infection, acute kidney injury and glomerular disease, and chronic kidney disease.

3

Obstruction & Kidney Stones

▼

When urine can't drain, pressure backs up into the kidney — and stones are the classic cause of sudden obstruction.

Concept	Mechanism
Hydronephrosis	Obstruction backs urine up → the renal pelvis/calyces **dilate** under pressure (e.g., ureteropelvic junction obstruction)
↓ GFR during obstruction	Back-pressure rises in the tubules, opposing filtration → **filtration falls**; prolonged obstruction can cause AKI
↑ infection risk	Stagnant, retained urine is a breeding ground → more UTIs
Postobstructive diuresis	After relief, **retained fluid and solutes are dumped** → large urine output (watch for dehydration/electrolyte loss)
Kidney stones (calculi)	**Concentrated urine** (dehydration) + supersaturation → crystals; **calcium oxalate** is the most common type

Key idea — the obstruction chain: blockage → **back-pressure** → **hydronephrosis** + **falling GFR** + **stasis/infection**. Relieve it and you may see **postobstructive diuresis** as the kidney unloads retained fluid and solutes.

Pearl — renal colic (Carlos's case): a stone causes **sudden severe flank pain radiating to the groin**, in **waves (colicky)**, with restlessness, nausea, and **concentrated urine**. Staying hydrated (dilute urine) helps prevent calcium oxalate stones.

PRACTICE

Q1. What process causes hydronephrosis?

A. Obstruction backs urine up, dilating the renal pelvis and calyces under pressure

B. Increased GFR

C. Excess water intake

D. Loss of renal blood flow only

A is correct. Blocked outflow raises pressure that distends the collecting system — hydronephrosis (e.g., from a ureteropelvic junction obstruction).

PRACTICE

Q2. Which process explains postobstructive diuresis after relief of an obstruction?

A. Retained fluid and solutes are excreted once flow is restored

B. The kidney stops making urine

C. Aldosterone surges to retain water

D. GFR falls permanently

A is correct. After the blockage clears, accumulated fluid and solutes are dumped, producing a large diuresis — monitor for dehydration and electrolyte loss.

PRACTICE

Q3. Which condition promotes kidney stone formation?

A. Concentrated urine (dehydration) causing supersaturation of stone-forming salts

B. Dilute urine

C. High fluid intake

D. Low urine calcium

A is correct. Concentrated urine supersaturates with calcium oxalate (the most common stone), allowing crystals to form and grow.

4 Urinary Tract Infection ▼

Most UTIs are **ascending** — bacteria travel **up** from the urethra. The higher they go, the more serious.

Level	What it is	Clues
Cystitis (lower UTI)	Bladder infection/inflammation	**Urgency, frequency, dysuria** (burning), suprapubic discomfort
Pyelonephritis (upper UTI)	Infection reaches the **kidney** (usually ascending from the bladder)	**Flank pain, fever, chills**, systemic illness

Key idea: UTIs are usually **ascending** (urethra → bladder → up the ureter to the kidney). **Cystitis** = bladder (urgency/frequency/burning). **Pyelonephritis** = kidney (flank pain + fever). **Stasis** from obstruction or reflux makes infection more likely.

Pearl: the irritated bladder wall in **cystitis** signals the urge to void even when nearly empty → the classic **urgency and frequency**. Anything that traps urine (obstruction, incomplete emptying, reflux) raises infection risk.

PRACTICE

Q1. Which mechanism explains an ascending urinary tract infection?

A. Bacteria travel up from the urethra into the bladder and toward the kidney

B. Bacteria spread from the blood only

C. The kidney filters bacteria into urine

D. Bacteria descend from the lungs

A is correct. Most UTIs ascend from the urethra; if they reach the kidney they cause pyelonephritis.

PRACTICE

Q2. What causes urgency and frequency with cystitis?

A. Bladder wall inflammation triggers the urge to void even when not full

B. Increased GFR

C. Reduced bladder sensation

D. Ureteral dilation

A is correct. Inflammation irritates the bladder, producing a frequent, urgent need to urinate plus burning (dysuria).

5 Acute Kidney Injury & Glomerular Disease ▼

Acute kidney injury (AKI) is a sudden drop in GFR. **Glomerular disease** damages the filter itself, leaking what should stay in the blood.

AKI — sudden loss of filtration

- **Prerenal** (most common): poor blood flow to the kidney (**dehydration**, hemorrhage, shock) → ↓ **GFR** → **oliguria** (low urine output).
- **Intrarenal:** direct kidney damage (toxins, ischemia, myoglobin).
- **Postrenal:** obstruction (Topic 3).
- Falling GFR → **creatinine and waste rise**; urine output drops.

Glomerular disease — a leaky filter

- **Glomerulonephritis**: inflammation damages the glomerulus → **blood leaks into urine (hematuria)**.
- **Nephrotic pattern**: barrier damage → massive **protein loss (proteinuria)** → low blood albumin → **edema**.

> **Key idea: dehydration → prerenal AKI → oliguria** (the kidney is starved of blood, so GFR and urine fall, and creatinine rises). When the **glomerular filter** itself is injured, it leaks: inflammation → **blood (hematuria)**; barrier breakdown → **protein (proteinuria)** → **edema**.

PRACTICE

Q1. A patient with severe dehydration develops oliguria. Which mechanism explains it?

A. Reduced renal blood flow lowers GFR, so less urine is formed (prerenal AKI)

B. Increased GFR

C. Excess ADH suppression

D. Glomerular protein loss

A is correct. Dehydration drops blood flow to the kidney → GFR falls → oliguria (low output) — the classic prerenal AKI mechanism.

PRACTICE

Q2. What explains hematuria in glomerular disease?

A. Inflammation damages the glomerular filter, letting red blood cells leak into the urine

B. Bladder infection

C. A kidney stone scraping the ureter

D. Increased urine concentration

A is correct. A damaged, inflamed glomerulus becomes leaky to red blood cells → hematuria (blood in urine).

PRACTICE

Q3. What change decreases urine output in acute kidney injury?

A. A fall in glomerular filtration rate

B. A rise in GFR

C. Increased renal perfusion

D. Decreased creatinine

A is correct. AKI lowers GFR, so less filtrate is formed and urine output drops (oliguria), while creatinine and wastes accumulate.

6 Chronic Kidney Disease ▼

Chronic kidney disease (CKD) is progressive, irreversible loss of nephrons over time. As GFR declines, the kidney fails at **excretion, fluid/electrolyte balance, and hormone production**.

Consequence	Mechanism
Most common cause	**Diabetes** (then hypertension) damages the small renal vessels over years
Anemia (fatigue, pallor)	Damaged kidneys make less **erythropoietin (EPO)** → fewer RBCs
Hyperkalemia	Failing kidneys can't excrete **potassium** → it builds up (dangerous for the heart)
Fluid overload / acidosis / waste buildup	Lost excretory and regulatory capacity → edema, metabolic acidosis, rising creatinine/BUN
Bone & mineral changes	Altered calcium/phosphate and vitamin D handling

Key idea: CKD's **#1 cause is diabetes**. Two classic patho links to remember: **low EPO → anemia** (fatigue, pallor) and **failure to excrete potassium → hyperkalemia** (the electrolyte CKD patients are at greatest risk for).

PRACTICE

Q1. Which condition is the most common cause of chronic kidney disease?

A. Diabetes mellitus

B. Kidney stones

C. A single urinary tract infection

D. Dehydration

A is correct. Diabetes (followed by hypertension) damages the kidney's small vessels over time and is the leading cause of CKD.

PRACTICE

Q2. A patient with CKD develops fatigue and pallor. Which mechanism best explains these findings?

A. Reduced erythropoietin production lowers red blood cell production (anemia)

B. Increased EPO

C. Iron overload

D. Excess oxygen delivery

A is correct. Diseased kidneys make less EPO, so fewer RBCs are produced — anemia causes the fatigue and pallor.

PRACTICE

Q3. Which electrolyte imbalance is a CKD patient at greatest risk for developing?

A. Hyperkalemia (high potassium)

B. Hypokalemia

C. Hypernatremia only

D. Hypocalcemia only

A is correct. Failing kidneys can't excrete potassium well, so it accumulates — hyperkalemia, which is dangerous for cardiac rhythm.

SECTION 3

Pediatric Renal & Urinary

Congenital and obstructive problems, the childhood glomerular diseases, the renal tumor, and why infants are fluid-fragile.

7 Congential & Obstructive (Reflux, PKD)

Condition	Mechanism
Vesicoureteral reflux (VUR)	The valve where the ureter enters the bladder fails → urine flows **backward** toward the kidney → **recurrent infections** and scarring
Ureteropelvic junction (UPJ) obstruction	Narrowing where the renal pelvis meets the ureter → back-pressure → **hydronephrosis**
Congenital bladder outlet obstruction (e.g., posterior urethral valves)	Blockage at the bladder outlet → back-pressure up the tract → **hydronephrosis and kidney damage** if untreated
Polycystic kidney disease (PKD)	Numerous **fluid-filled cysts** enlarge and **compress/replace** functional kidney tissue → progressive loss of function

Key idea: normally a one-way valve (the vesicoureteral junction) **prevents urine refluxing** from bladder to ureter. In **VUR** that valve fails, so urine refluxes up and brings bacteria with it → **recurrent UTIs** and renal scarring. Obstructions anywhere (UPJ, bladder outlet) cause the same back-pressure picture: **hydronephrosis**.

Pearl — PKD: it isn't a blockage — it's **cysts** that grow and squeeze out working nephrons, so kidney function declines as the cysts expand.

PRACTICE

Q1. Which mechanism normally prevents reflux of urine from the bladder into the ureters?

A. A one-way valve at the ureterovesical (bladder) junction

B. The glomerular filter

C. The loop of Henle

D. The renal capsule

A is correct. The junction acts as a one-way valve; when it fails (VUR), urine refluxes toward the kidney → recurrent infections.

PRACTICE

Q2. What explains recurrent infections with vesicoureteral reflux?

A. Urine refluxes backward toward the kidney, carrying bacteria upward

B. The kidney over-filters bacteria

C. Increased bladder emptying

D. Reduced bladder pressure

A is correct. Backward urine flow moves bacteria from the bladder up to the kidneys, causing repeated UTIs and potential scarring.

PRACTICE

Q3. Which mechanism reduces kidney function in polycystic kidney disease?

A. Growing fluid-filled cysts compress and replace functional kidney tissue

B. A single obstructing stone

C. Bacterial infection only

D. Low erythropoietin

A is correct. Multiple enlarging cysts crowd out and destroy normal nephrons over time, progressively reducing function.

8

Pediatric Glomerular Disease & HUS

▼

Condition	Mechanism
Poststreptococcal glomerulonephritis	**Immune complexes** form after a strep infection and deposit in the glomeruli → inflammation → **hematuria** (and edema, hypertension); appears **after the infection resolves**
Nephrotic syndrome (childhood)	Glomerular barrier damage → heavy **proteinuria** → low blood albumin → **edema** (low oncotic pressure)
Hemolytic uremic syndrome (HUS)	Often after toxin-producing *E. coli* → **endothelial injury & microthrombi** in the kidney → renal injury (with hemolytic anemia + low platelets)

Key idea — "nephritic vs nephrotic": inflammation that lets **blood** through = **hematuria** (poststrep glomerulonephritis — Aiden's case, kidney injury appearing *after* the strep throat clears). Barrier breakdown that lets **protein** out = **proteinuria** → **edema** (nephrotic syndrome). **HUS** injures the kidney through **microthrombi/endothelial damage**.

Pearl — Aiden's case: untreated strep pharyngitis → immune complexes deposit in the glomeruli → **poststreptococcal glomerulonephritis**, which shows up *after* the sore throat resolves with **hematuria** (tea/cola-colored urine), edema, and high blood pressure.

PRACTICE

Q1. What causes hematuria after poststreptococcal glomerulonephritis?

A. Immune complexes deposit in the glomeruli, inflaming and damaging the filter

B. A bladder stone

C. Direct bacterial filtration

D. Increased erythropoietin

A is correct. Post-strep immune complexes lodge in the glomeruli and trigger inflammation, making the filter leaky to red blood cells → hematuria.

PRACTICE

Q2. What causes the edema in childhood nephrotic syndrome?

A. Heavy urinary protein loss lowers blood albumin and oncotic pressure, so fluid leaks into tissues

B. Excess albumin in the blood

C. Increased GFR

D. High potassium

A is correct. Losing protein in the urine drops plasma albumin, lowering oncotic pressure so fluid shifts into the tissues → edema.

PRACTICE

Q3. What causes renal injury in hemolytic uremic syndrome?

A. Endothelial damage and microthrombi obstruct small renal vessels

B. A blocked ureter

C. Low erythropoietin

D. Bacterial cystitis

A is correct. HUS (often after Shiga-toxin *E. coli*) injures small-vessel endothelium and forms microthrombi that damage the kidney — alongside hemolytic anemia and thrombocytopenia.

9

Wilms Tumor & Infant Fluid Risk

▼

Topic	Mechanism
Nephroblastoma (Wilms tumor)	Malignant proliferation of **embryonic kidney cells** → a (often large, one-sided) abdominal/renal mass in young children
Congenital bladder outlet obstruction (untreated)	Persistent back-pressure → hydronephrosis → **progressive kidney damage/failure**
Infant fluid fragility	Infants have a **higher body-water percentage, higher turnover, and immature kidneys** that concentrate urine poorly → they **dehydrate quickly** and are prone to fluid/electrolyte imbalance

Key idea: Wilms tumor (nephroblastoma) is the classic childhood kidney cancer — embryonic renal cells proliferating into a renal mass. And remember **why infants are fluid-fragile:** more total body water, faster turnover, and **immature kidneys** that can't concentrate urine well — so they lose fluid fast and tip into imbalance quickly.

PRACTICE

Q1. Which mechanism best explains nephroblastoma (Wilms tumor)?

A. Malignant proliferation of embryonic kidney cells forming a renal mass

B. Bacterial kidney infection

C. A blocking stone

D. Immune-complex deposition

A is correct. Wilms tumor arises from primitive (embryonic) kidney cells that proliferate into a mass — the most common childhood renal cancer.

PRACTICE

Q2. A newborn has severe congenital bladder outlet obstruction. Which complication is most likely if untreated?

A. Back-pressure causes hydronephrosis and progressive kidney damage

B. Spontaneous resolution with no risk

C. Increased GFR

D. Low potassium

> **A is correct.** Persistent outlet obstruction backs urine up the tract, distending the kidneys (hydronephrosis) and damaging them over time.

PRACTICE

Q3. Why are infants at higher risk for fluid imbalance?

A. Higher body-water percentage and turnover with immature kidneys that concentrate urine poorly

B. Lower body-water percentage than adults

C. Fully mature concentrating ability

D. Very slow fluid turnover

> **A is correct.** Infants have proportionally more water, faster turnover, and immature kidneys that can't concentrate urine well, so they dehydrate quickly.

EXAM

Renal & Urinary Cheat Sheet

The highest-yield patterns — review the night before.

Function

- **GFR** driven by glomerular **hydrostatic (blood) pressure**. **Proximal tubule** reabsorbs the most; **loop of Henle** (countercurrent) concentrates urine.

- **Creatinine ↑ = GFR ↓** (kidney function marker).
- **Low BP → renin → RAAS** (angiotensin II vasoconstriction + aldosterone Na/water retention) → BP up. **ADH** retains water. **Sympathetic** constricts afferent arteriole → ↓GFR.

Alterations

- **Obstruction** → back-pressure → **hydronephrosis** + ↓GFR + infection risk; relief → **postobstructive diuresis**. **Stones** = concentrated urine; **calcium oxalate** most common; renal colic = flank→groin pain in waves.
- **UTI** usually **ascending**: cystitis (bladder — urgency/frequency/dysuria) vs pyelonephritis (kidney — flank pain + fever).
- **AKI**: prerenal (dehydration/shock) → ↓GFR → **oliguria**, ↑ creatinine. **Glomerular disease**: inflammation → **hematuria**; barrier loss → **proteinuria** → **edema**.
- **CKD**: #1 cause **diabetes**; ↓EPO → **anemia**; can't excrete K^+ → **hyperkalemia**.

Pediatric

- **VUR** = failed bladder-ureter valve → reflux → recurrent UTIs. **UPJ / bladder-outlet obstruction** → hydronephrosis. **PKD** = cysts crowd out nephrons.
- **Poststrep glomerulonephritis** = immune complexes → hematuria (after strep resolves). **Nephrotic syndrome** = proteinuria → edema. **HUS** = endothelial injury/microthrombi (often E. coli).
- **Wilms tumor (nephroblastoma)** = embryonic kidney-cell tumor. **Infants** = more body water + immature kidneys → dehydrate fast.

 TERMS

Glossary

Quick definitions of the Week 11 vocabulary.

Nephron

The kidney's functional unit (glomerulus + tubule).

GFR

Glomerular filtration rate — volume of blood filtered per minute; the key measure of kidney function.

Creatinine

Waste cleared by filtration; rises when GFR falls.

RAAS

Renin-angiotensin-aldosterone system; raises blood pressure/volume via vasoconstriction and Na/water retention.

ADH

Antidiuretic hormone; promotes water reabsorption in the collecting duct.

Hydronephrosis

Dilation of the renal collecting system from obstructed urine flow.

Postobstructive diuresis

Large urine output after relieving an obstruction as retained fluid/solutes are excreted.

Renal calculi (stones)

Crystallized deposits (commonly calcium oxalate) forming in concentrated urine.

Cystitis / Pyelonephritis

Bladder infection / kidney infection (usually ascending).

Acute kidney injury (AKI)

Sudden drop in GFR; prerenal, intrarenal, or postrenal.

Oliguria

Abnormally low urine output.

Hematuria / Proteinuria

Blood / protein in the urine from glomerular damage.

Chronic kidney disease (CKD)

Progressive, irreversible nephron loss; leading cause is diabetes.

Vesicoureteral reflux (VUR)

Backward urine flow from bladder to ureter from a failed valve → recurrent UTIs.

Polycystic kidney disease

Cysts that enlarge and replace functional kidney tissue.

Poststreptococcal glomerulonephritis

Immune-complex glomerular inflammation after strep → hematuria.

Nephrotic syndrome

Heavy proteinuria → low albumin → edema.

Hemolytic uremic syndrome (HUS)

Endothelial injury/microthrombi (often E. coli) → renal injury, anemia, low platelets.

Nephroblastoma (Wilms tumor)

Childhood kidney cancer from embryonic renal cells.